

Sampling Distributions and Common Random Variables

Technical reference

Companion to the note on lorenzoliuzzo.github.io. Standard random variables and the distributions of sample statistics, with derivations.

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1 Discrete random variables

A discrete random variable has mass function $p(x) = P(X = x)$ on a countable support, with $\sum_x p(x) = 1$, $E[X] = \sum_x xp(x)$ and $\text{Var}(X) = E[X^2] - (E[X])^2$. Probability background is in the Probability Foundations reference.

1.1 Bernoulli and binomial

Definition. $X \sim \text{Bernoulli}(p)$ if $P(X = 1) = p$ and $P(X = 0) = 1 - p$. $Y \sim \text{Binomial}(n, p)$ is the sum of n independent $\text{Bernoulli}(p)$ variables, with mass function $P(Y = k) = \binom{n}{k} p^k (1 - p)^{n-k}$, $k = 0, \dots, n$.

Theorem (Mean and variance). If $X \sim \text{Bernoulli}(p)$ then $E[X] = p$ and $\text{Var}(X) = p(1 - p)$. If $Y \sim \text{Binomial}(n, p)$ then $E[Y] = np$ and $\text{Var}(Y) = np(1 - p)$.

Proof. $E[X] = 1 \cdot p + 0 \cdot (1 - p) = p$, and since $X^2 = X$, $E[X^2] = p$ and $\text{Var}(X) = p - p^2$. For $Y = \sum_{i=1}^n X_i$ with independent X_i , linearity gives $E[Y] = np$ and independence gives $\text{Var}(Y) = \sum_i \text{Var}(X_i) = np(1 - p)$. $\square$

The binomial mass function counts the $\binom{n}{k}$ arrangements of k successes among n trials, each with probability $p^k (1 - p)^{n-k}$. For large n it is approximately $\mathcal{N}(np, np(1 - p))$ by the central limit theorem; a common rule is $np(1 - p) \geq 10$.

1.2 Geometric

Definition. $X \sim \text{Geometric}(p)$ counts the trials up to and including the first success: $P(X = k) = (1 - p)^{k-1}p$, $k = 1, 2, \dots$

Theorem (Mean and memorylessness). $E[X] = 1/p$ and $\text{Var}(X) = (1 - p)/p^2$. Moreover $P(X > m + k \mid X > m) = P(X > k)$ for all integers $m, k \geq 0$.

Proof. With $q = 1 - p$, $E[X] = p \sum_{k \geq 1} kq^{k-1} = p/(1 - q)^2 = 1/p$, using $\sum_{k \geq 1} kq^{k-1} = \frac{d}{dq} \sum_{k \geq 0} q^k = (1 - q)^{-2}$. A second differentiation gives $E[X(X - 1)]$ and hence the variance. For the tail, $P(X > k) = q^k$, so $P(X > m + k \mid X > m) = q^{m+k}/q^m = q^k$. $\square$

1.3 Poisson

Definition. $X \sim \text{Poisson}(\lambda)$ has $P(X = k) = e^{-\lambda} \lambda^k / k!$, $k = 0, 1, 2, \dots$

Theorem (Moments, sums and the binomial limit). (i) $E[X] = \text{Var}(X) = \lambda$. (ii) If $X \sim \text{Poisson}(\lambda)$ and $Y \sim \text{Poisson}(\mu)$ are independent then $X + Y \sim \text{Poisson}(\lambda + \mu)$. (iii) If $X_n \sim \text{Binomial}(n, \lambda/n)$ then $P(X_n = k) \rightarrow e^{-\lambda} \lambda^k / k!$ for every fixed k .

Proof. (i) $E[X] = \sum_{k \geq 1} k e^{-\lambda} \lambda^k / k! = \lambda e^{-\lambda} \sum_{k \geq 1} \lambda^{k-1} / (k-1)! = \lambda$, and similarly $E[X(X - 1)] = \lambda^2$, so $\text{Var}(X) = \lambda^2 + \lambda - \lambda^2 = \lambda$. (ii) $P(X + Y = m) = \sum_{j=0}^m e^{-\lambda} \lambda^j / j! \cdot e^{-\mu} \mu^{m-j} / (m - j)! = e^{-(\lambda + \mu)} (\lambda + \mu)^m / m!$ by the binomial theorem. (iii) $P(X_n = k) = [n! / ((n - k)! n^k)] \cdot (\lambda^k / k!) (1 - \lambda/n)^{n-k}$. As $n \rightarrow \infty$ the first factor tends to 1, $(1 - \lambda/n)^n \rightarrow e^{-\lambda}$ and $(1 - \lambda/n)^{-k} \rightarrow 1$. $\square$

2 Continuous random variables

A continuous random variable has a density f with $P(a \leq X \leq b) = \int_a^b f$, mean $E[X] = \int x f(x) dx$ and distribution function $F(x) = \int_{-\infty}^x f$.

2.1 Uniform and inverse transform sampling

$X \sim \text{Uniform}(a, b)$ has $f(x) = 1/(b - a)$ on $[a, b]$, $E[X] = (a + b)/2$ and $\text{Var}(X) = (b - a)^2/12$ (compute $E[X^2] = (a^2 + ab + b^2)/3$).

Theorem (Inverse transform). Let F be a distribution function, $F^{-1}(u) = \inf\{x : F(x) \geq u\}$, and $U \sim \text{Uniform}(0, 1)$. Then $F^{-1}(U)$ has distribution function F . Conversely, if X has a continuous distribution function F , then $F(X) \sim \text{Uniform}(0, 1)$.

Proof. The generalized inverse satisfies $F^{-1}(u) \leq x$ iff $u \leq F(x)$, so $P(F^{-1}(U) \leq x) = P(U \leq F(x)) = F(x)$. For the converse, with F continuous and strictly increasing on the support, $P(F(X) \leq u) = P(X \leq F^{-1}(u)) = F(F^{-1}(u)) = u$. $\square$

The converse is the probability integral transform used for the uniformity of p -values; the direct statement is how simulation software turns uniform random numbers into draws from any distribution with a computable F^{-1} .

2.2 Exponential

Definition. $X \sim \text{Exponential}(\lambda)$ has $f(x) = \lambda e^{-\lambda x}$, $x \geq 0$, so that $P(X > x) = e^{-\lambda x}$.

Theorem (Moments and memorylessness). $E[X] = 1/\lambda$, $\text{Var}(X) = 1/\lambda^2$, and $P(X > s + t \mid X > s) = P(X > t)$.

Proof. Integrating by parts, $E[X] = \int_0^\infty x\lambda e^{-\lambda x} dx = 1/\lambda$ and $E[X^2] = 2/\lambda^2$, so $\text{Var}(X) = 1/\lambda^2$. For the tail, $P(X > s + t \mid X > s) = e^{-\lambda(s+t)}/e^{-\lambda s} = e^{-\lambda t}$. $\square$

The exponential is the only continuous distribution on $[0, \infty)$ with this property. If events arrive as a Poisson process of rate λ (the count in an interval of length t is $\text{Poisson}(\lambda t)$), the waiting time to the first event, and between successive events, is $\text{Exponential}(\lambda)$: $P(T > t) = P(\text{no event in } [0, t]) = e^{-\lambda t}$.

2.3 Normal

Definition. $X \sim \mathcal{N}(\mu, \sigma^2)$ has density $(2\pi\sigma^2)^{-1/2} \exp(-(x - \mu)^2/(2\sigma^2))$. The standard normal $\mathcal{N}(0, 1)$ has distribution function Φ and quantiles $z_p = \Phi^{-1}(p)$.

Theorem (Standardization and closure). (i) $E[X] = \mu$ and $\text{Var}(X) = \sigma^2$. (ii) $Z = (X - \mu)/\sigma \sim \mathcal{N}(0, 1)$, so $P(X \leq x) = \Phi((x - \mu)/\sigma)$. (iii) If $X_1, \dots, X_n$ are independent, $X_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$, then $\sum_i a_i X_i + b \sim \mathcal{N}(\sum_i a_i \mu_i + b, \sum_i a_i^2 \sigma_i^2)$.

Proof. (ii) The substitution $y = (u - \mu)/\sigma$ in $\int_{-\infty}^x f(u) du$ gives $\int_{-\infty}^{(x-\mu)/\sigma} \varphi(y) dy$. (i) follows from (ii) since $E[Z] = 0$ (odd integrand) and $E[Z^2] = 1$ (integration by parts). (iii) The mean and variance follow from linearity and independence; that the sum is again normal follows from the moment generating function $E[e^{tX}] = e^{\mu t + \sigma^2 t^2/2}$, which multiplies for independent summands. $\square$

Quantiles that recur: $z_{0.90} \approx 1.282$, $z_{0.95} \approx 1.645$, $z_{0.975} \approx 1.960$, $z_{0.995} \approx 2.576$, and $P(|Z| \leq 1, 2, 3) \approx 0.683, 0.954, 0.997$.

2.4 Gamma and chi-square

Definition. $X \sim \text{Gamma}(\alpha, \beta)$ (shape $\alpha > 0$, rate $\beta > 0$) has density $\beta^\alpha x^{\alpha-1} e^{-\beta x} / \Gamma(\alpha)$ on $x > 0$, with $E[X] = \alpha/\beta$ and $\text{Var}(X) = \alpha/\beta^2$. The sum of k independent $\text{Exponential}(\lambda)$ variables is $\text{Gamma}(k, \lambda)$. The chi-square distribution is $\chi_k^2 = \text{Gamma}(k/2, 1/2)$, with density $x^{k/2-1} e^{-x/2} / (2^{k/2} \Gamma(k/2))$.

Theorem (Chi-square as a sum of squares). If $Z_1, \dots, Z_k$ are independent $\mathcal{N}(0, 1)$ then $\sum_i Z_i^2 \sim \chi_k^2$, with mean k and variance $2k$.

Proof. The density of Z_1^2 follows by the change of variable $y = z^2$ and equals the χ_1^2 density; independent Gamma variables with a common rate add their shapes, giving χ_k^2 . For the moments, $E[Z^2] = 1$ and $\text{Var}(Z^2) = E[Z^4] - 1 = 3 - 1 = 2$, and the Z_i^2 are independent, so the mean is k and the variance $2k$. $\square$

3 Sampling distributions of statistics

A statistic $T = T(X_1, \dots, X_n)$ is a random variable. Its sampling distribution is the law of T when the sample is drawn from the assumed model, and its standard error is $\text{se}(T) = \sqrt{\text{Var}(T)}$.

3.1 Mean, variance and proportion

Theorem (Mean and variance of $\bar{X}$). For i.i.d. X_i with mean μ and variance σ^2 , $E[\bar{X}] = \mu$ and $\text{Var}(\bar{X}) = \sigma^2/n$. For two independent samples, $\bar{X}_1 - \bar{X}_2$ has mean $\mu_1 - \mu_2$ and variance $\sigma_1^2/n_1 + \sigma_2^2/n_2$.

Proof. Linearity gives $E[\bar{X}] = n^{-1} \sum_i E[X_i] = \mu$; independence gives $\text{Var}(\bar{X}) = n^{-2} \sum_i \text{Var}(X_i) = \sigma^2/n$. For the difference, $\text{Var}(\bar{X}_1 - \bar{X}_2) = \text{Var}(\bar{X}_1) + \text{Var}(\bar{X}_2)$ by independence. $\square$

If the data are normal, $\bar{X} \sim \mathcal{N}(\mu, \sigma^2/n)$ exactly; in general the central limit theorem gives $\sqrt{n}(\bar{X} - \mu)/\sigma \xrightarrow{d} \mathcal{N}(0, 1)$.

Theorem (Sample proportion). If $Y \sim \text{Binomial}(n, p)$ and $\hat{p} = Y/n$, then $E[\hat{p}] = p$, $\text{Var}(\hat{p}) = p(1-p)/n$, and $\sqrt{n}(\hat{p} - p)/\sqrt{p(1-p)} \xrightarrow{d} \mathcal{N}(0, 1)$.

Proof. The moments follow from those of the binomial and the scaling rule $\text{Var}(Y/n) = \text{Var}(Y)/n^2$. The limit is the central limit theorem applied to $Y = \sum_i X_i$ with Bernoulli summands (the de Moivre–Laplace theorem). $\square$

Example (Classifier accuracy). A classifier with true accuracy $p = 0.8$ evaluated on $n = 100$ independent test examples has $\text{se}(\hat{p}) = \sqrt{0.8 \cdot 0.2/100} = 0.04$, so the observed accuracy lies in 0.8 ± 0.08 with probability about 95%.

3.2 Normal-theory distributions

Definition (Gaussian family). $X \sim \mathcal{N}(\mu, \sigma^2)$ has density $(2\pi\sigma^2)^{-1/2} \exp(-(x - \mu)^2/(2\sigma^2))$. Let $Z_1, \dots, Z_k \sim \mathcal{N}(0, 1)$ be independent. Then

- $\sum_i Z_i^2 \sim \chi_k^2$ (chi-square, k degrees of freedom), with mean k and variance $2k$;
- if $Z \sim \mathcal{N}(0, 1)$ and $V \sim \chi_k^2$ are independent, $Z/\sqrt{V/k} \sim t_k$ (Student);
- if $V_1 \sim \chi_a^2$ and $V_2 \sim \chi_b^2$ are independent, $(V_1/a)/(V_2/b) \sim F_{a,b}$.

Facts used below: a linear combination of jointly Gaussian variables is Gaussian; for jointly Gaussian variables, uncorrelated implies independent; $t_k \rightarrow \mathcal{N}(0, 1)$ as $k \rightarrow \infty$; $t_k^2 \sim F_{1,k}$.

Let $X_1, \dots, X_n \sim \mathcal{N}(\mu, \sigma^2)$ be i.i.d., with $\bar{X} = \bar{X}_n$ and $S^2 = (n-1)^{-1} \sum_i (X_i - \bar{X})^2$.

Theorem (Distribution of $\bar{X}$ and S^2). (i) $\bar{X} \sim \mathcal{N}(\mu, \sigma^2/n)$. (ii) $\bar{X}$ and S^2 are independent. (iii) $(n-1)S^2/\sigma^2 \sim \chi_{n-1}^2$. (iv) Consequently

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}. \quad (1)$$

Proof. (i) is closure of the Gaussian family under linear maps. For (ii)–(iii), let $Z_i = (X_i - \mu)/\sigma$ and Q an orthogonal $n \times n$ matrix whose first row is $(1/\sqrt{n}, \dots, 1/\sqrt{n})$. Then $Y = QZ$ has $Y_1 = \sqrt{n}\bar{Z}$ and, since Q is orthogonal, $Y \sim \mathcal{N}(0, I_n)$: the Y_i are independent standard normals. Orthogonality preserves the norm: $\sum_i Z_i^2 = \sum_i Y_i^2$, and $\sum_i Z_i^2 = \sum_i (Z_i - \bar{Z})^2 + n\bar{Z}^2$, hence

$$\sum_i (Z_i - \bar{Z})^2 = \sum_{i=2}^n Y_i^2 \sim \chi_{n-1}^2, \quad (2)$$

a function of $(Y_2, \dots, Y_n)$ only, and so independent of Y_1 and thus of $\bar{Z}$. Since $\sum_i (Z_i - \bar{Z})^2 = (n-1)S^2/\sigma^2$ and $\bar{X} = \mu + \sigma\bar{Z}$, (ii) and (iii) follow. For (iv), write $T = [(\bar{X} - \mu)/(\sigma/\sqrt{n})] / \sqrt{((n-1)S^2/\sigma^2)/(n-1)}$, a standard normal over the root of an independent $\chi_{n-1}^2/(n-1)$. $\square$

Theorem (Unbiasedness of S^2). For any i.i.d. sample with finite variance σ^2 , $E[S^2] = \sigma^2$.

Proof. $\sum_i (X_i - \bar{X})^2 = \sum_i (X_i - \mu)^2 - n(\bar{X} - \mu)^2$. Taking expectations, $E[\sum_i (X_i - \bar{X})^2] = n\sigma^2 - n(\sigma^2/n) = (n-1)\sigma^2$. $\square$

3.3 The t and F distributions in more detail

The t_k distribution has density $\Gamma((k+1)/2)/(\sqrt{k\pi}\Gamma(k/2)) \cdot (1+t^2/k)^{-(k+1)/2}$. It is symmetric about 0, has mean 0 for $k > 1$ and variance $k/(k-2)$ for $k > 2$, and has heavier tails than $\mathcal{N}(0, 1)$, to which it converges as $k \rightarrow \infty$. Selected 97.5% quantiles are $t_{5,0.975} = 2.571$, $t_{9,0.975} = 2.262$, $t_{29,0.975} = 2.045$, against $z_{0.975} = 1.960$.

Lemma (Relation between t and F). If $T \sim t_k$ then $T^2 \sim F_{1,k}$.

Proof. Write $T = Z/\sqrt{V/k}$ with $Z \sim \mathcal{N}(0, 1)$ independent of $V \sim \chi_k^2$. Then $T^2 = Z^2/(V/k) = (Z^2/1)/(V/k)$ with $Z^2 \sim \chi_1^2$, the ratio defining $F_{1,k}$. $\square$

The $F_{a,b}$ distribution has mean $b/(b-2)$ for $b > 2$ and is the null distribution of the ratio of two independent normal-sample variance estimates, S_1^2/S_2^2 scaled by σ_2^2/σ_1^2 , with $a = n_1 - 1$ and $b = n_2 - 1$; it is also the null distribution of the overall test in analysis of variance and linear regression.

3.4 A non-normal example: the sample maximum

Theorem (Maximum of uniform samples). Let $X_1, \dots, X_n$ be i.i.d. $\text{Uniform}(0, \theta)$ and $M = \max_i X_i$. Then $P(M \leq m) = (m/\theta)^n$ for $0 \leq m \leq \theta$, with density nm^{n-1}/θ^n , $E[M] = n\theta/(n+1)$, and $(n+1)M/n$ is unbiased for θ .

Proof. $M \leq m$ iff every $X_i \leq m$, so by independence $P(M \leq m) = (m/\theta)^n$. Differentiating gives the density, and $E[M] = \int_0^\theta m \cdot nm^{n-1}/\theta^n dm = n\theta/(n+1)$. $\square$

The sampling distribution of M is concentrated against the boundary θ and is far from normal, so normal-theory intervals do not apply: sampling distributions must be derived for the statistic actually used.

3.5 Simulation and the bootstrap

Any sampling distribution can be approximated by simulation: draw B samples from the assumed model, compute T on each, and use the empirical distribution of $T_1, \dots, T_B$; by the law of large numbers its mean and standard deviation converge to $E[T]$ and $\text{se}(T)$ as $B \rightarrow \infty$.

When the model is unknown, the bootstrap replaces the unknown distribution F by the empirical distribution $\hat{F}_n$, which puts mass $1/n$ on each observation. Resample n observations with replacement, compute T_b^* , repeat for $b = 1, \dots, B$, and estimate $\text{se}(T)$ by the standard deviation of the T_b^* . It is consistent for smooth statistics such as the mean; it fails for extreme statistics such as the maximum in the previous example.

4 References

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